PHARMACOLOGY · MCQ PRACTICE

Emetic and Antiemetic Drugs

LG 2
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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. A patient asks which neurotransmitter receptors are involved in triggering vomiting. Which four receptor types does the lecture name? EASY

A. Serotonin (5-HT3), acetylcholine (muscarinic), dopamine (D2), and histamine (H1)
B. GABA, glutamate, glycine, and adenosine receptors in the gut wall
C. Opioid, cannabinoid, NK1, and nicotinic acetylcholine
D. Beta-adrenergic, alpha-adrenergic, angiotensin, and serotonin
E. Only dopamine and serotonin receptors

2. A patient is given ondansetron for postoperative nausea. Which class of antiemetic is this drug? EASY

A. An NK1 receptor antagonist
B. A 5-HT3 receptor antagonist
C. A muscarinic receptor antagonist
D. A dopamine D2 receptor antagonist
E. A cannabinoid

3. A patient is prescribed hyoscine (scopolamine) for motion sickness prophylaxis. What type of receptor does this drug block? EASY

A. Dopamine D2 receptors
B. Histamine H1 receptors
C. Muscarinic receptors
D. 5-HT3 receptors
E. NK1 receptors

4. A patient on nabilone for chemotherapy-induced nausea asks what drug class it belongs to. What is the correct answer? EASY

A. A benzodiazepine
B. A glucocorticoid
C. A phenothiazine
D. A cannabinoid
E. An aminoglycoside

5. A patient receiving radiotherapy is given granisetron. Where is this drug's primary site of action? MEDIUM

A. The vestibular nuclei of the inner ear
B. The higher cortical centres of the brain
C. The vomiting centre in the medulla directly
D. Vagal afferents in the gut wall
E. The chemoreceptor trigger zone (CTZ)

6. A patient taking palonosetron asks why it is dosed much less frequently than ondansetron. What distinguishes it pharmacokinetically? MEDIUM

A. Its tissue half-life is around 40 hours, far longer than the 4–9 hours of other 5-HT3 antagonists
B. It is not metabolized at all by any organ and is excreted entirely unchanged within minutes of dosing
C. It has 100% first-pass hepatic extraction, prolonging its action
D. It permanently and irreversibly binds the 5-HT3 receptor
E. It is only effective when combined with dexamethasone

7. A patient with renal impairment needs a 5-HT3 antagonist. What does the lecture say about dose adjustment in this situation, compared to liver impairment? MEDIUM

A. Dose reduction is needed in renal impairment, but not in liver impairment at any stage
B. No dose reduction is needed in renal impairment, but dose reduction is needed in liver impairment
C. Dose reduction is needed in both renal and liver impairment regardless of severity
D. No dose reduction is needed in either renal or liver impairment under any circumstances
E. 5-HT3 antagonists are contraindicated in any degree of renal impairment

8. A patient on metoclopramide for gastro-oesophageal reflux develops galactorrhoea and gynecomastia. What explains this adverse effect? MEDIUM

A. Metoclopramide directly stimulates oestrogen receptors in breast tissue
B. Metoclopramide blocks 5-HT3 receptors in the pituitary gland
C. Metoclopramide's dopamine blockade stimulates prolactin release
D. Metoclopramide induces hepatic enzymes that increase prolactin synthesis
E. Metoclopramide causes fluid retention that mimics breast changes

9. A patient is prescribed aprepitant for chemotherapy-induced emesis and is also taking clarithromycin. What interaction should be anticipated? MEDIUM

A. No interaction is expected at all, since aprepitant is not metabolized by the liver in any meaningful way
B. Clarithromycin will have no effect because aprepitant is a prodrug activated only in the gut lumen
C. Clarithromycin will decrease aprepitant levels substantially by inducing its hepatic metabolism
D. Clarithromycin, a strong CYP3A4 inhibitor, should be avoided with aprepitant since aprepitant is metabolized by this enzyme
E. Clarithromycin interacts only with fosaprepitant, never with oral aprepitant itself

10. A patient with a history of cardiac arrhythmia needs an antiemetic for cytotoxic-drug-induced vomiting. Which 5-HT3 antagonist carries the most notable QT-prolongation risk? MEDIUM

A. Granisetron
B. Ondansetron
C. Palonosetron
D. Tropisetron
E. Dolasetron

11. A patient receiving diphenhydramine alongside high-dose metoclopramide asks why both are given. What is the purpose of adding the antihistamine? MEDIUM

A. To reduce extrapyramidal reactions caused by high-dose metoclopramide
B. To increase the sedative effect for better sleep during chemotherapy
C. To directly potentiate the antiemetic action of metoclopramide at the CTZ
D. To treat the cardiac arrhythmia risk from metoclopramide
E. To prevent metoclopramide-induced hepatotoxicity

12. A patient is prescribed dexamethasone as part of an antiemetic regimen for cytotoxic chemotherapy. What is stated about its mechanism? MEDIUM

A. It directly and selectively blocks 5-HT3 receptors located in the CTZ
B. The mechanism is not clear, but it may involve blockade of prostaglandins
C. It is a competitive antagonist at dopamine D2 receptors
D. It selectively blocks NK1 receptors in the vomiting centre
E. It works purely through sedation, reducing anxiety-related nausea

13. A patient receiving emetogenic chemotherapy is given a combination of dexamethasone, diphenhydramine, droperidol, and metoclopramide. According to the bar chart in the lecture, roughly what response rate (%) did this four-drug combination achieve against emetic episodes in the first 24 hours after cisplatin? MEDIUM

A. 91%
B. 63%
C. 76%
D. 58%
E. 100%

14. A patient taking oral metoclopramide notes it must be taken fairly frequently through the day. What dosing interval does the lecture give, and why does the drug undergo extensive first-pass handling? MEDIUM

A. 5 mg once daily, reflecting its very long half-life
B. 40 mg every 12 hours, reflecting its poor oral bioavailability
C. 10–20 mg every 6 hours, because it is not metabolized at all
D. 10–20 mg every 6 hours, reflecting its high first-pass metabolism
E. 10–20 mg every 6 hours, because it is excreted entirely unchanged in bile

15. A patient with respiratory failure accidentally receives apomorphine as an emetic. Why is this drug contraindicated in this setting? HARD

A. Apomorphine directly and irreversibly paralyzes the diaphragm muscle fibres
B. Apomorphine causes severe, life-threatening bronchospasm via H1 receptor activation
C. Apomorphine is metabolized to a toxic metabolite that permanently damages lung tissue
D. Apomorphine blocks central chemoreceptors that are needed for normal breathing regulation
E. Apomorphine has a CNS depressant effect, which is dangerous in respiratory failure

16. A child ingests kerosene and the caregiver considers giving ipecacuanha to induce vomiting. Why is this contraindicated? HARD

A. Ipecacuanha is contraindicated in kerosene and corrosive poisoning, listed alongside unconscious patients and morphine poisoning
B. Ipecacuanha is contraindicated only in children under two years of age, with no other listed exceptions
C. Ipecacuanha has absolutely no emetic effect whatsoever in any case of poisoning
D. Ipecacuanha is contraindicated only when it is given together with activated charcoal
E. Ipecacuanha is contraindicated simply because its onset of action takes over an hour

17. A patient taking rolapitant for chemotherapy-induced nausea is also started on a drug metabolized by CYP2D6. What interaction should be anticipated, distinct from aprepitant's profile? HARD

A. Rolapitant has no CYP interactions whatsoever at any dose, quite unlike aprepitant
B. Rolapitant is a moderate inhibitor of CYP2D6, which could raise levels of CYP2D6 substrates
C. Rolapitant is instead a strong inducer of CYP2D6, which lowers substrate levels considerably
D. Rolapitant and aprepitant are described as sharing an identical CYP2D6 interaction profile
E. Rolapitant interacts only with CYP3A4 substrates and never with CYP2D6 substrates at all

18. A patient needs an NK1 antagonist for chemotherapy but cannot swallow oral medication. Which drug in this class is specifically described as administered intravenously, and why? HARD

A. Aprepitant, because it is only available as an IV formulation
B. Rolapitant, because it cannot be absorbed orally at all
C. Fosaprepitant, because it is a prodrug of aprepitant given IV
D. Netupitant, because it is inactivated by gastric acid
E. Substance P itself, because it must bypass hepatic metabolism

19. A patient with Ménière's disease has recurrent vomiting from vestibular disorders. Which H1 antagonist does the clinical-uses slide name specifically for vestibular disorders like Ménière's disease? HARD

A. Promethazine
B. Hyoscine
C. Nabilone
D. Cinnarizine
E. Prochlorperazine

20. A patient's antiemetic effect from nabilone is reversed after naloxone administration. What does this observation suggest about nabilone's mechanism, per the lecture? HARD

A. That nabilone is itself an opioid receptor agonist that has no cannabinoid activity at all
B. That naloxone works by directly and selectively blocking CB1 cannabinoid receptors
C. That nabilone's antiemetic effect is entirely unrelated to the CTZ in every respect
D. That this interaction is clinically irrelevant and remains totally unexplained in pharmacology
E. That opioid receptors may be important in nabilone's antiemetic mechanism of action

✓ Answer key

Q	Answer	Difficulty	Why
1	**A**	EASY	The lecture lists serotonin (5-HT3), acetylcholine (muscarinic), dopamine (D2), and histamine (H1) as the neurotransmitters involved in vomiting.
2	**B**	EASY	Ondansetron is a 5-HT3 receptor antagonist, along with granisetron, dolasetron, palonosetron, and tropisetron.
3	**C**	EASY	Hyoscine (scopolamine) is a muscarinic receptor antagonist, used for prophylaxis and treatment of motion sickness.
4	**D**	EASY	Nabilone is a synthetic cannabinoid used to decrease vomiting from CTZ-stimulating agents.
5	**E**	MEDIUM	5-HT3 receptor antagonists act primarily at the CTZ, the main site for sensing emetic stimuli.
6	**A**	MEDIUM	5-HT3 antagonists generally have a tissue half-life of 4–9 hours, but palonosetron's half-life is about 40 hours.
7	**B**	MEDIUM	5-HT3 antagonists are excreted by kidney and liver; no dose reduction is needed in renal impairment, but it is needed in liver impairment.
8	**C**	MEDIUM	Metoclopramide stimulates prolactin release (via dopamine blockade), causing galactorrhoea, menstrual disorders, impotence, and gynecomastia.
9	**D**	MEDIUM	Aprepitant and rolapitant undergo hepatic metabolism primarily by CYP3A4; co-administration with strong CYP3A4 inhibitors (e.g. clarithromycin, ketoconazole) or inducers (e.g. rifampicin, phenytoin) should be avoided.
10	**E**	MEDIUM	QT interval prolongation among 5-HT3 antagonists is noted as more common with dolasetron.
11	**A**	MEDIUM	Antihistamines such as diphenhydramine are combined with high-dose metoclopramide to reduce extrapyramidal reactions, or with corticosteroids to counter metoclopramide-induced diarrhea.
12	**B**	MEDIUM	High-dose glucocorticoids (particularly dexamethasone) control emesis by an unclear mechanism that may involve blockade of prostaglandins.
13	**C**	MEDIUM	The dexamethasone + diphenhydramine + droperidol + metoclopramide combination showed about 76% response, behind dexamethasone + ondansetron (91%).
14	**D**	MEDIUM	Metoclopramide is given 10–20 mg every 6 hours; it undergoes high first-pass metabolism and is excreted in urine and breast milk.
15	**E**	HARD	Apomorphine has a CNS depressant effect, so it is contraindicated in respiratory failure.
16	**A**	HARD	The lecture lists kerosene and corrosive poisoning, unconscious patients, and morphine poisoning as contraindications for ipecacuanha.
17	**B**	HARD	Rolapitant is a moderate inhibitor of CYP2D6 specifically, while aprepitant is noted as an inducer of CYP3A4 and CYP2C9 and a dose-dependent inhibitor of CYP3A4 — different profiles.
18	**C**	HARD	Fosaprepitant is a prodrug of aprepitant, administered intravenously, while aprepitant itself is given orally.
19	**D**	HARD	Cinnarizine is named for motion sickness and vestibular disorders such as Ménière's disease, while promethazine is specifically named for severe morning sickness of pregnancy and cyclizine for postoperative/narcotic-related nausea.
20	**E**	HARD	The lecture states nabilone's antiemetic effect is antagonised by naloxone, implying opioid receptors may be important in its mechanism.